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1. What is optical flow in computer vision?

1 / 1 point

- ☐ A type of camera lens
- ☒ A method for estimating motion in an image
- ☐ A technique for changing the focus of a camera
- ☐ A way to enhance image contrast

✔ Correct

Optical flow in computer vision is the perceived motion of objects or points in an image or a sequence of images over time. It represents how objects appear to move from one frame to another in a video or a sequence of images.

2. Which of the following is NOT a common challenge in optical flow estimation?

1 / 1 point

- ☐ Illumination changes
- ☐ Occlusions
- ☒ Slow-moving objects
- ☐ Non-rigid deformations

✔ Correct

Challenges in optical flow estimation typically include factors like illumination changes, occlusions (objects blocking others), and non-rigid deformations (objects changing shape). Slow-moving objects are generally easier to track accurately because their motion is less ambiguous.

3. In the context of optical flow, the motion field describes:

1 / 1 point

- ☐ The motion of a single pixel in an image
- ☐ The overall motion of the camera
- ☒ The motion of objects in a scene
- ☐ The distortion of an image due to lens aberrations

✔ Correct

The motion field in optical flow analysis describes how objects in a scene are moving over time. It is a representation of the apparent motion of objects or features within the image or video sequence. Each point in the image corresponds to a specific motion vector that indicates the perceived movement of that point.